



A Study of Icon Design Styles for Recognition in HUD Environments

Dihui Chu
School of Mechanical Engineering,
Southeast University
China
220214976@seu.edu.cn

Zhisheng Zhang*
School of Mechanical Engineering,
Southeast University
China
oldbc@seu.edu.cn

Fangzhou Dong
Department of Art, Design and
Architecture, Monash University
Australia
Fangzhou.Dong@monash.edu

Wenyu Wu
School of Mechanical Engineering,
Southeast University
China
wuwenyu.design@seu.edu.cn

Yunlong Tang
Department of Mechanical and
Aerospace Engineering, Monash
University
Australia
Yunlong.Tang1@monash.edu

Abstract

In field of AR research and icon design, a problem worthy of further discussion is whether the design style change will affect the user's cognitive efficiency. With the development of intelligent automobile, the display of icons is no longer limited to the mobile phone and computer, but extends to the intelligent cockpits. The icon design style inside the human-computer interaction interface in automobile HMI and HUD directly relates to the driver's cognitive performance and driving safety. Especially in the HUD, due to the vibration of the car itself, the display effect of the icon is significantly different from the traditional display interface. Therefore, the design style of the icon may have different effects on the driver's cognition in different driving environments. In order to investigate this issue in depth, this paper designed a visual search task through simulation experiments to compare the effects of three different design styles of icons (linear icons, filled icons, and combined linear-filled Icon) on drivers' recognition speeds in different vibration directions (horizontal and vertical) and at three vibration frequency levels (5 Hz, 10 Hz, and 15 Hz). The experimental results show that the linear design style icons display particularly well in the vibration environment, while in the vibration direction, the icons with transverse vibration are more recognizable compared to those with longitudinal vibration. These findings provide a reference for optimizing the design of icons in in-vehicle HUDs, which can help to improve drivers' cognitive efficiency and driving safety.

*Corresponding author

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CCS Concepts

• Human computer interaction (HCI); • HCI design and evaluation methods; • Usability testing;

Keywords

Human factors, 3D UI design, Head-up Display, vibration

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1 Background

As the pivotal interface for contemporary in-vehicle interactions, the Heads-Up Display (HUD) revolutionizes the experience by replacing physical instrument clusters, fostering unparalleled convenience. The seamless integration of HUD enables drivers to seamlessly access vital vehicle and navigation data without the need to avert their gaze, thereby alleviating driver anxiety [1]. Furthermore, it enhances the overall driving experience through its immersive nature [2], offering a level of engagement that transcends traditional interfaces.

The HUD's capacity to present visual information tailored to human cognitive patterns and characteristics is instrumental in mitigating driving fatigue, a cornerstone of safe motoring. Consequently, delving into the human factors that arise from the interplay between the driver and the HUD interface holds immense potential for significantly enhancing driving safety. By conducting research in this realm, we can effectively optimize the HUD's design to better serve the needs of drivers, ultimately promoting safer and more enjoyable journeys.

Drawing upon Wang's seminal research on cognitive load and HUD-related theory [3], this paper meticulously outlines an association model that intertwines cognitive load, drivers' visual task activities, and key HUD design elements, as depicted in Figure 1. These design elements are comprehensively categorized into interface layout, information structure, interaction mode, and visual

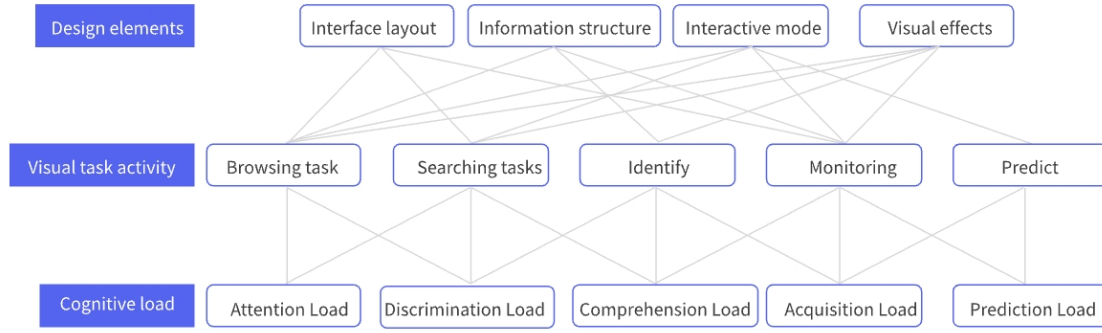


Figure 1: Correlation of Cognitive Load, Task Activity, Design Elements and Task Activity

effect. The driver’s visual tasks, encompassing browsing, searching, recognition, monitoring, and prediction, are intricately linked, particularly with regards to visual effect, underscoring its pivotal role in modulating cognitive load.

Optimizing the visual effect holds immense potential in effectively regulating drivers’ cognitive load. This encompasses the strategic design of text, graphics, shapes, colors, animations, and more. Existing studies have delved into the effectiveness of straightforward textual information, illustrative patterns, metaphorical color schemes, and attention-grabbing animations, all of which have proven to captivate drivers’ attention, clarify priorities, and expedite recognition.

In this paper, we narrow our focus on the design style of icons, embarking on an exploration of how the display effect influences drivers’ recognition abilities within a vibrational environment. By delving into this specific aspect of HUD design, we aim to contribute to the ongoing dialogue on enhancing driver safety and experience through innovative interface solutions.

2 Icons In the Driving Environment

2.1 Effects of vibration on visual perception

Different from the fixed information display in the past, HUD is gradually applied to more dynamic scenes, such as moving vehicles, ships and airplanes, where drivers can obtain driving status information and environmental information from the HUD in front of them without looking down, which greatly improves driving safety. In the HUD display, information appears in the form of text (speed, RPM, mileage, fuel level) and images (alert signs, warning signs). Clear information display and clear information communication are very important for drivers, however, many existing design principles are based on design guidelines that are only applicable to stable and static environments, and the vibration of the HUD display in a moving vehicle can place an additional burden on the visual system, for example, the human eye feels motion blur when reading text in a dynamic environment. Lewis and Griffin [4] found that vibration had a significant effect on visual cognition regardless of condition by having subjects sit in a vibrating seat with and without pedals for a digital reading task. In 1986, Lewis and Griffin [5] built on

the original experiment by comparing different distances and vibration frequencies and found that when subjects were sitting at a distance less than 1.5 meters from the object being observed, the effect of vibration increased with decreasing observation distance, and the effect of vibration at 3.15 Hz became more influential with decreasing distance compared to 16 Hz. At observation distances greater than 1.5 meters, the effect of vibration on observation was independent of distance. The subject’s viewing angle vibrates while the monitor remains stationary, then reasonable visual acuity is maintained at about 8 Hz, and if only the monitor vibrates while the subject does not move, then visual discrimination begins to deteriorate at frequencies greater than 0.6 Hz. In terms of image information, Andersson [6] found in a study on a military HUD that the complexity of symbols had a significant effect on visual performance. The orientation of the symbols is also important for their readability. Symbols consisting of horizontal lines would be difficult to recognize during vertical vibration. Li [7] verified that the line width, font, and size of text in AR environments all affect human cognition, and that the color contrast and luminance contrast of the HUD information also affects human visual searcher task performance.

According to the dual coding theory proposed by Paivio [8], the human perceptual system consists of two coding modalities, Verbal Code and Nonverbal Code. Verbal Code responsible for processing logogen information, and Nonverbal Code responsible for processing imagen information. Logogen is the coded representation unit of information perceived in the form of language, organized according to associations and hierarchies; imagen is the coded representation unit of nonverbal things and events and situations, organized in the relationship between parts and the whole. Logogen elements are the coded representations of information perceived in the form of language, organized according to associations and hierarchies; imagen elements are the coded representations of non-verbal things, events, and situations, organized in the relationship between parts and the whole. For example, a person can visualize an elephant by using the word “elephant”, or he can mentally visualize an elephant by looking at a picture of an elephant and then describe it in words. For the human perception mode, no matter it is structural information or unstructured information, the

free recall effect of imagen information is better, and the burden of cognitive load is less, so it is important to conduct more in-depth research on the icon display information of HUD.

This paper intends to enrich the human factors related research in HUD experiments by exploring the reaction performance of people in vibrating environments on HUD pixel information, starting from the effect of vibration in HUD on people's cognitive performance.

2.2 Icon design style

The outline of the icon determines people's perception, using the outline to define the boundaries of the object, perceive the shape, and the shape of the icon is determined by the design style, so the design style of the icon is an important element for users to perceive and obtain information, and can make a quick judgment. Icons can be categorized in various ways, according to the style can be divided into pictogram icons, text icons and abstract icons, according to the artistic approach can be divided into anthropomorphic icons and flattened icons, according to the use of the device environment can be divided into PC icons, mobile icons, smart wearable device icons and so on. In the static environment, researchers usually tend to start from the perspective of visual complexity [9], and study the effects of the number of icons, strokes, lines, line widths and other factors on cognitive performance. According to Feature Integration Theory [10], simple icons are easier to understand than complex icons, and complex icons take longer to understand, but in special environments, complex icons may be more accurate and efficient in conveying visual information. Hsuan Lin [11] once compared the effects of face icons and linear icons on visual attention in static environments, and found that flat icons are more attractive. In this paper, we started from the classification of linear icons, flat icons, and line-flat combination icons, and takes into consideration of HUD application environments, combines the direction of vibration and the frequency of vibration, and explores the interaction effect between different design styles, vibration directions, and vibration frequencies, so as to provide design style suggestions for the design of automotive HUDs and aviation HUD icons.

3 Experiment design

3.1 Equipment and environment

'Razer Blade 13' laptop, with a resolution of 1920*1080, 13.3 inches, a brightness of 423 cd/m², and a control room brightness of 500 lx, the experimental environment was well insulated with uniform illumination and no environmental interference. Eprime 3.0 was used to present the cued stimuli, Adobe After Effect was used to edit the pseudo-randomized target stimulus video and icon vibration effects, and Hololens was used to play the video of the experimental material.

3.2 Experiment Subjects

Twenty subjects participated in this experiment (11 males and 9 females), all of whom were master's degree students at Southeast University, with the age range of the subjects ranging from 23-26, and all of whom did not have color blindness or color deficiency and had normal corrected visual acuity (over 1.0).

3.3 Experiment material Design

In this paper, the object of research is the vibration of the driving environment in the pixel information, select the icon as the experimental material is more illustrative, so the commonly used icons in the automotive interactive interface were selected, the design styles were linear, filled, combined linear and filled icons, each group of icons in the shape of a high degree of proximity to the semantic meaning of icons in different design styles is the same, not due to the change in design styles lead to semantic ambiguity, the experimental paradigm in this study was improved based on the visual search paradigm. In the context of Hololens, black and transparency are perceived subjectively as indistinguishable, necessitating a design approach that prevents icons from merging into the background environment. Consequently, the experimental stimulus icon design eschews the use of black hues and instead selects icon colors that exhibit a pronounced contrast with the surrounding environment, ensuring that the icons remain distinct and easily discernible. Currently for the HUD depth of field design is fixed, for the HUD depth of field design of the numerical value is not a unified standard, according to research of Li, K. [12], previous scholars have suggested that the symbol text is placed at a distance of about 2 meters from the user's eyes, there is also a proposal to be placed at 4-8 meters, the aviation field placed in the depth of field of the HUD design is usually located in the infinite distance. The ground transportation community generally agrees that the focus of the display should be between 2.0-2.5 meters. The current mainstream HUD focal design was adopted in this experiment, with the distance of the icon imaging surface set to approximately 2.5m from the subject.

3.4 Experiment variables

The design styles of the icons in the experiment were divided into three kinds, each design style totaled 49, presenting a 7 × 7 combination arrangement, according to the human eye on the visual sensitivity range study, the effect of vibration on the human eye perception and visual performance is mainly manifested in the low-frequency interval from 2 to 20 Hz, with a sensitivity amplitude of 1.66 mm, in this paper, we set the amplitude frequency of the amplitude of the three levels of 5Hz, 10Hz and 20Hz, the amplitude is set at 1.66, the vibration direction is horizontal, vertical and random. Therefore, this experiment was designed with a total of: 3 icon styles × 3 vibration frequencies × 3 directions=27 experimental combinations, with each case occurring randomly 4 times, for a total of 108 trials.

3.5 Experimental process

Ensure that the subjects are well rested to maintain a good mental state before the experiment, followed by an explanation and a practice experiment, subjects will be informed of the experimental procedures and operational requirements, the distance between the subject's eyes and the screen is about 70cm, the screen in Hololens is adjusted to a distance of 2.5m away from the subject's eye position, after each set of experiments the subjects will be given a 10-minute break to allow the eyes to remain relaxed, to avoid visual transient effect.

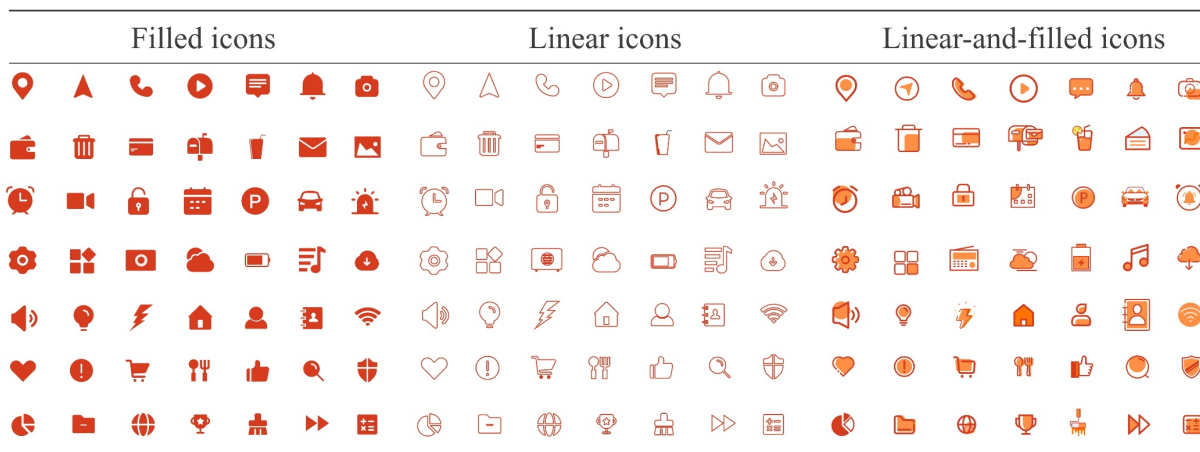


Figure 2: linear icons, filled icons, and linear-and-filled icons

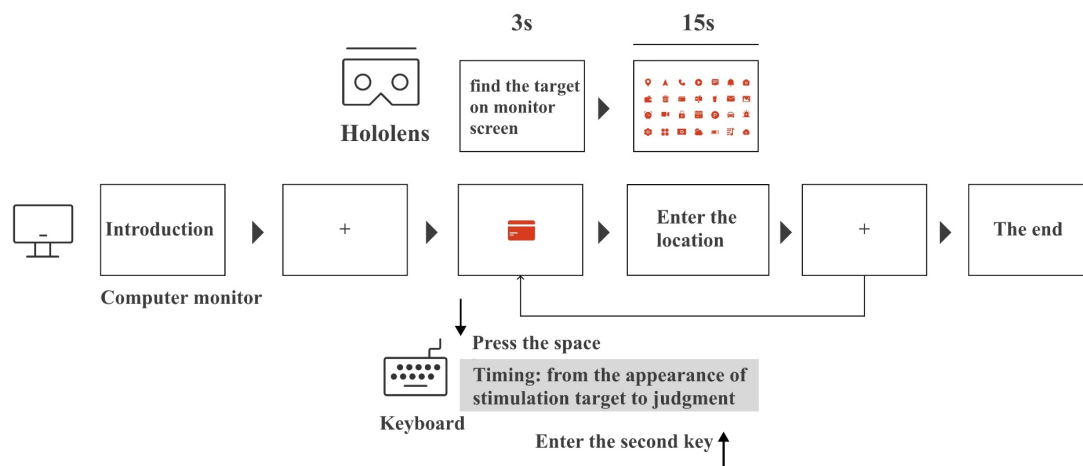


Figure 3: Experimental flow chart



Figure 4: Diagrams of the experimental scene and what the subjects see

Table 1: Results of variance analysis

Term	Sum of squares	Degree of freedom	Mean Square	F	P	R ²	Ad-justed R ²
Intercept	55008.546	1	55008.546	95996.743	<0.001***	0.685	0.669
Icon design style	88.278	2	44.139	77.028	<0.001***		
Vibration direction	155.542	2	77.771	135.72	<0.001***		
Vibration frequency	346.435	2	173.218	302.286	<0.001***		
Icon design style	48.869	20	2.443	4.264	<0.001***		
Vibration direction							
Vibration frequency							
Error	293.962	513	0.573		NaN		

Note: ***, **, * represent 1 %, 5 % and 10 % significance levels, respectively.

In the experiment, subjects were required to perform a judgment task by combining observational information from pictures presented in the computer monitor and videos played in Hololens. After reading the introduction to the experiment and pressing space, a “+” blank screen focus would appear, and then a pseudo-randomized target stimulus video would start to play in Hololens, with one trial presented every 18 s. The first 3 s of the video for each trial prompted “find the target on monitor screen”, the subject needed to remember the target stimulus from the computer monitor and determine which design style icon it was. In the next 15 s, Hololens would present a picture of a randomized combination of icons, with each icon not repeating, and the subject needed to turn their attention away from the monitor to find the target stimulus in Hololens. Hololens to find the location of the target stimulus, and press the keyboard to record the location of the target stimulus.

Eprime records the reaction time for each judgment, and subjects are required to press the corresponding keypad according to the position of the target stimulus in the chaotic combination and the design style, which corresponds to “F” for filled icons, “J” for linear, “L” for linear-and-filled, and “L” for linear-and-filled icons. “L”. For example, if the target stimulus was a solid icon, which appeared in the second row of the first line, subjects needed to press the numeric keys “F”, “1”, “2”, and the timing started from the time the target stimulus appeared by pressing the space to the time the third numeric key was pressed to stop the timing, and the timing stopped after the third numeric key was pressed. After pressing the third numeric key to stop the timing, subjects waited for the cue to appear after judging one trial and then continued to press the space for the next trial judgment, if the judgment was not completed in time, then press the space twice to skip and wait for the next trial judgment. Each experiment took about 35 minutes to conduct.

4 Experiment results

After organizing the experimental data, extreme values brought about by random variables and unstable disturbances in the subjects’ physiology and psychology were eliminated from the results, and after three-factor ANOVA, it could be seen that different design styles vibration frequency and vibration direction both had significant effects on reaction time ($P < 0.01$), and there was a main effect. After Pearson correlation analysis, it can be seen that there

is a positive correlation between vibration direction and vibration frequency, between vibration frequency and reaction time, between vibration direction and reaction time, and between design style and reaction time.

There is a positive correlation between vibration frequency and reaction time. From the graph, it can be seen that the lowest reaction time occurs at vibration frequency 5Hz, and the longest reaction time occurs at 15Hz, which indicating with the vibration frequency increasing, more subjects’ cognitive resources were occupied. In terms of vibration direction, the vibration recognition performance of horizontal icons is better than that of vertical vibrations, while the readability of random directional vibrations is the worst. Overall, in vibration environments, linear icons have the best judgment performance, followed by filled icons, and linear-and-filled combined icons have the longest reaction time. Combined with two factor analysis, linear icons have the best recognition ability in lateral vibration. Linear icons still maintain the best recognition ability in vibration frequencies of 5Hz and 10Hz, while linear and filled icons have strong recognition ability at 15Hz vibration frequencies.

5 Limitation and future work

The experiment in this paper is inspired by Gabbard’s monocular AR visual search task [13], and the experimental design is improved in this study. Hololens used in this paper better restored the driver’s observation effect in the binocular environment, provides the stereoscopic parallax cue during binocular observation, and restores the driver’s real observation experience.

The environment of Hololens is AR information, and the computer display is real environment information. Just as the driver needs to constantly switch attention between AR environment and real environment when driving a vehicle, using Hololens as an experimental tool is more realistic than developing a vibration simulation interface program on a computer display. However, the application of Hololens itself will bring several new problems. One is that the weight of AR glasses is an additional burden for most subjects. Some subjects reported that the wearing of Hololens caused additional fatigue after the experiment, which may interfere with visual judgment. Secondly, the difference between Holons’ brightness and fineness and computer monitors may also lead to additional burdens, and the frequent transfer of participants’

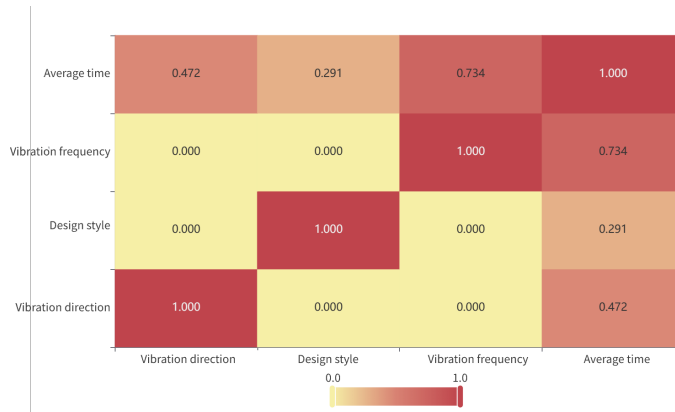


Figure 5: Correlation coefficient heat map

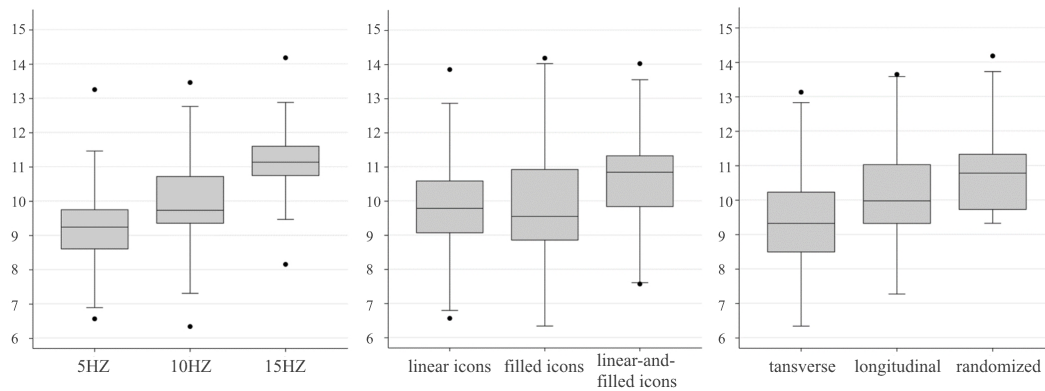


Figure 6: Reaction time of different vibration frequency, different design style and different vibration direction

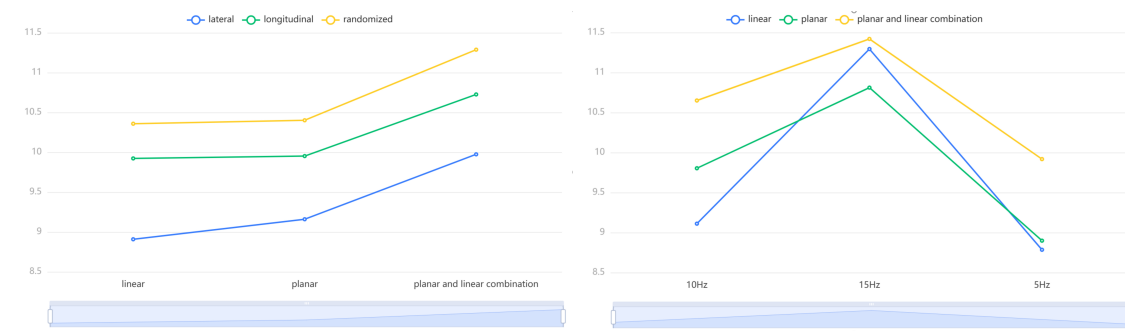


Figure 7: factor analysis diagram of design style and vibration direction and vibration frequency and designstyle

vision from the AR environment to the real environment may lead to fatigue. Moreover, the virtual image in Hololens is actually farther than the users' focus point, and the user has the possibility of convergence conflict, which also brings potential interference possibilities for the scene of restoring HUD in Hololens. In future work, we can combine driving simulation to explore the icon vibration identification of driving tasks in a dynamic environment and use the visual search task as a secondary task to characterize the

influence of reading the vibration icon on the driving task during driving.

In the design of the experimental material in this paper, in order to distinguish the color from the background environment, the linear icon and the filled icon both use red (#D73B20), and the linear-and-filled icons combination icon uses a certain proportion of orange (#FF9045). The influence of color on the identification of icons in the vibration environment has not been discussed. In the

future, more in-depth work can further study the cognitive effects of color and environment integration, contrast and brightness in the vibration environment, and study their interaction effects.

6 Conclusion

This paper reports the driver's recognition efficiency with icons in different design styles (filled shape, linear and linear-and-filled combined icons) in HUD under lateral vibration, longitudinal vibration and random vibration direction and three-level design vibration frequency (5Hz, 10Hz and 15Hz). The experiment restores the AR-HUD display effect in binocular vision environment through Hololens, which enabled a more realistic simulation of the binocular vision environment encountered by drivers using AR-HUD. The experimental conclusion is that, in general, based on the influence of vibration, linear icons are more suitable for the display environment of AR-HUD, while linear-and-filled icons will cause great difficulties to the driver's recognition, and compared with longitudinal vibration, lateral vibration has less influence on the driver's recognition. In terms of vibration frequency, the icon recognition of all design styles will decrease with the increase in vibration frequency. This study provides a reference basis for the selection of icon design styles in the HUD environment, and by selecting icons with better visual recognition effects in the vibration environment can effectively improve driver recognition efficiency and enhance driving safety.

References

- [1] A. Lanatà *et al.*, "How the Autonomic Nervous System and Driving Style Change With Incremental Stressing Conditions During Simulated Driving," *IEEE Transactions on Intelligent Transportation Systems*, vol. 16, no. 3, pp. 1505–1517, 2015, doi: 10.1109/TITS.2014.2365681.
- [2] C. Fan and S. He, "Micromirror-based real image laser automotive head-up display," *International Journal of Optomechatronics*, vol. 11, no. 1, pp. 15–26, 2017, doi: 10.1080/15599612.2017.1342884.
- [3] Y. Wang, "Study of Vehicle HUD Interactive Interface Design Based on Cognitive Load," Master's Thesis, Chongqing University, Chongqing, 2021.
- [4] C. H. Lewis and M. J. Griffin, "Predicting the effects of vibration frequency and axis, and seating conditions on the reading of numeric displays," *Ergonomics*, vol. 23, pp. 485–499, 1980.
- [5] M. J. Moseley and M. J. Griffin, "Effects of display vibration and whole-body vibration on visual performance," *Ergonomics*, vol. 29, no. 8, pp. 977–983, Aug. 1986, doi: 10.1080/00140138608967211.
- [6] P. Andersson and C. von Hofsten, "Readability of vertically vibrating aircraft displays," *Displays*, vol. 20, no. 1, pp. 23–30, 1999, doi: [https://doi.org/10.1016/S0141-9382\(98\)00056-0](https://doi.org/10.1016/S0141-9382(98)00056-0).
- [7] Y. Li, "Color hierarchy and application of hud interface," Master's Thesis, Southeast University, Nanjing, 2020.
- [8] A. Paivio, "Bilingual Dual Coding Theory and Memory," in *Foundations of Bilingual Memory*, R. R. Heredia and J. Altarriba, Eds., New York, NY: Springer New York, 2014, pp. 41–62. doi: 10.1007/978-1-4614-9218-4_3.
- [10] S. McDougall, I. Reppa, J. Kulik, and A. Taylor, "What makes icons appealing? The role of processing fluency in predicting icon appeal in different task contexts," *Applied Ergonomics*, vol. 55, pp. 156–172, 2016, doi: <https://doi.org/10.1016/j.apergo.2016.02.006>.
- [11] H. Lin, Y.-C. Hsieh, and F.-G. Wu, "A study on the relationships between different presentation modes of graphical icons and users' attention," *Comput. Hum. Behav.*, vol. 63, pp. 218–228, 2016.
- [12] K. Li, L. Bai, Y. Li, and Z. Zhou, "Distortion correction algorithm of ar-hud virtual image based on neural network model of spatial continuous mapping," in *2020 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, IEEE, 2020, pp. 178–183.
- [13] J. L. Gabbard, G. M. Fitch, and H. Kim, "Behind the Glass: Driver Challenges and Opportunities for AR Automotive Applications," *Proceedings of the IEEE*, vol. 102, no. 2, pp. 124–136, 2014.